ASSIGNMENT (LINEAR REGRESSION)

Question 1 : What is Simple Linear Regression (SLR)? Explain its purpose.

ANSWER-Simple Linear Regression is a statistical method used to model the relationship between two variables:

- One independent variable (X)
- One dependent variable (Y) It assumes that the relationship between X and Y can be represented by a straight line.

$$Y=a+bX$$

Question 2: What are the key assumptions of Simple Linear Regression?

Ans:- • Linearity

There is a **linear relationship** between the independent variable (X) and the dependent variable (Y).

 Change in X leads to a proportional change in Y.

- **Independence of Errors**

The residuals (errors) are **independent of each other**.

 One observation's error should not affect another's.

- **Homoscedasticity (Constant Variance)**

The variance of residuals is **constant across all values of X**.

 Errors should be evenly spread, not increasing or decreasing.

- **Normality of Errors**

The residuals should be **normally distributed**.

 This is especially important for hypothesis testing and confidence intervals.

- **No Multicollinearity (Automatically satisfied in SLR)**

Since SLR has only **one independent variable**, multicollinearity does not occur.

- **No Significant Outliers**

Extreme values should not heavily influence the regression line.

Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

ANS:- $Y=\beta 0+\beta 1X+\varepsilon$

- **Y (Dependent Variable)**

The output or response variable that we want to **predict or explain**.

- **X (Independent Variable)**

The input or predictor variable used to **predict Y**.

- **β_0 (Intercept)**

The value of **Y when X = 0**.
It shows where the regression line crosses the Y-axis.

- **β_1 (Slope / Regression Coefficient)**

Represents the **change in Y for a one-unit change in X**.
It indicates the **strength and direction** of the relationship.

- **ε (Error Term)**

The random error representing the **difference between actual and predicted values** of Y.
It accounts for factors not included in the model.

Question 4: Provide a real-world example where simple linear regression can be applied.

ANS:- A real-world example of Simple Linear Regression is predicting house prices based on the size of the house.

In this example:

- The **independent variable (X)** is the *area of the house* (in square feet).
- The **dependent variable (Y)** is the *price of the house*.

Simple Linear Regression can be used to analyze how the price of a house changes with an increase in its area. By collecting past data on house sizes and their corresponding prices, a regression line can be fitted to the data. This line helps in understanding the relationship between house size and price and allows us to predict the price of a house for a given area.

If the regression model shows a positive slope, it indicates that as the area of the house increases, the price also increases. Real estate companies and buyers can use this model to estimate house prices, plan investments, and make informed decisions.

Question 5: What is the method of least squares in linear regression?

ANS:- The Method of Least Squares is a statistical technique used in linear regression to determine the best-fitting regression line for a given set of data points.

In Simple Linear Regression, the regression line is represented by the equation:

Y=β0+β1XY = \beta_0 + \beta_1 XY=β0+β1X

The method of least squares works by minimizing the sum of the squares of the residuals.
A residual is the difference between the actual observed value of the dependent variable and the value predicted by the regression model.

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

ANS:- Logistic Regression is a statistical and machine learning technique used to model the relationship between one or more independent variables and a binary dependent variable. The dependent variable in logistic regression takes only two possible outcomes, such as 0 or 1, Yes or No, or Pass or Fail.

Instead of predicting continuous values, logistic regression predicts the probability that a given input belongs to a particular class. It uses a logistic (sigmoid) function to transform the linear combination of input variables into a value between 0 and 1.

Linear Regression is used when the dependent variable is continuous, while Logistic Regression is used when the dependent variable is categorical. Logistic Regression is especially useful for classification problems where the outcome needs to be predicted as a probability.

Question 7: Name and briefly describe three common evaluation metrics for regression models.

ANS:- Evaluation metrics are used to measure the **performance and accuracy** of regression models by comparing predicted values with actual values. Three commonly used evaluation metrics for regression models are:

1. Mean Absolute Error (MAE)

- Mean Absolute Error measures the **average absolute difference** between the actual values and the predicted values.
- It provides a clear interpretation of error in the same units as the dependent variable.

2. Mean Squared Error (MSE)

- Mean Squared Error calculates the **average of the squared differences** between actual and predicted values.
- It penalizes larger errors more heavily and is useful when large deviations are undesirable.

3. Root Mean Squared Error (RMSE)

- Root Mean Squared Error is the **square root of MSE**.
- It expresses error in the same units as the dependent variable and gives more weight to larger errors.

Question 8: What is the purpose of the R-squared metric in regression analysis?

ANS:- R-squared (R^2) is a statistical metric used to measure how well a regression model fits the observed data. It indicates the proportion of the variance in the dependent variable (Y) that is explained by the independent variable(s) (X).

The value of R^2 ranges from **0 to 1**:

- **$R^2 = 1$** → The model perfectly explains all the variation in Y.
- **$R^2 = 0$** → The model does not explain any variation in Y.

Purpose of R-squared:

1. **Assess model fit:**
   R^2 shows how much of the variation in the dependent variable is captured by the model.
2. **Compare models:**
   It allows comparison of different regression models; a higher R^2 indicates a better fit.
3. **Explain variance:**
   It quantifies the explanatory power of the independent variable(s) in predicting the dependent variable.

Question 9: Write Python code to fit a simple linear regression model using scikit-learn and print the slope and intercept.

(Include your Python code and output in the code box below.)

ANS:-

```
from sklearn.linear_model import LinearRegression
import numpy as np

# Sample data: X = independent variable, Y = dependent variable
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)  # Reshape for sklearn
Y = np.array([2, 4, 5, 4, 5])

# Create a Linear Regression model
model = LinearRegression()

# Fit the model to the data
model.fit(X, Y)

# Print the slope (coefficient) and intercept
print("Slope (β1):", model.coef_[0])
print("Intercept (β0):", model.intercept_)
```

```
OUTPUT:-
Slope (β1): 0.6
Intercept (β0): 2.2
```

Question 10: How do you interpret the coefficients in a simple linear regression model?

ANSWER:-

In a Simple Linear Regression model, the equation looks like this:

$$Y=\beta_0+\beta_1 X$$

Here, β_0 is the intercept and β_1 is the slope.

1. Intercept (β_0):

- This tells you the value of Y when X is 0.
- In other words, it's where the line starts on the Y-axis.
- **Example:** If $\beta_0 = 2$, it means that when X = 0, Y is predicted to be 2.

2. Slope (β_1):

- This tells you how much Y changes when X increases by 1.
- It shows the direction and strength of the relationship.
    - Positive slope: Y goes up when X goes up.
    - Negative slope: Y goes down when X goes up.
- Example: If $\beta_1 = 0.5$, then for every 1-unit increase in X, Y increases by 0.5 on average.